

Quan Huynh

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COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

Department of Chemical Engineering

University of Illinois Chicago

Chicago, Illinois 60607-7052

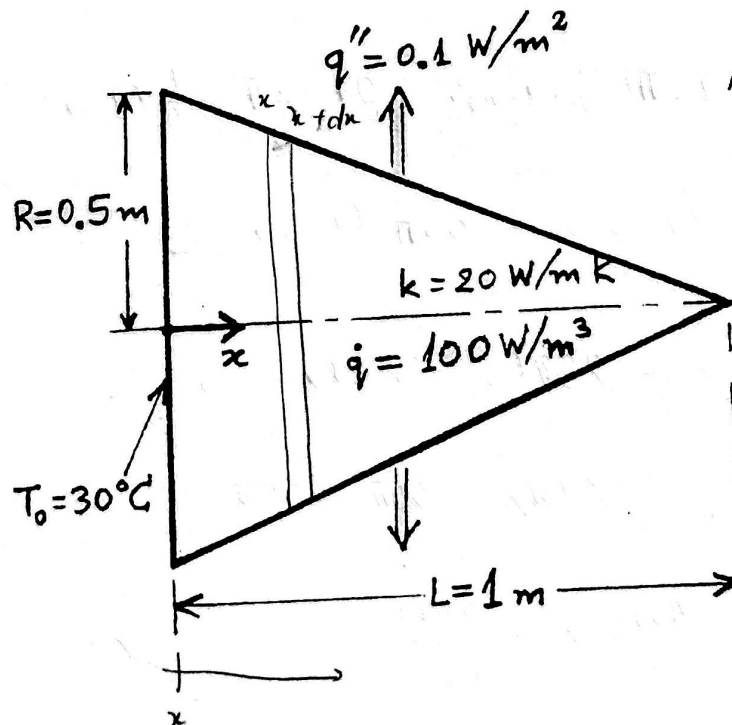
Quiz #2

September 30, 2025

Open Notes and Book

Duration: 40 minutes

A solid cone as shown is of circular cross section and the radius of its base is 0.5 m. The cone of length 1 m and thermal conductivity 20 W/m·K has uniform volumetric heat generation of 100 W/m³. A constant heat flux of 0.1 W/m² is leaving the cone through the lateral surface of the cone, while the end surface at $x = 0$ is maintained at $T_0 = 30^\circ\text{C}$. Determine the temperature profile $T(x)$ and the temperature $T(L)$.



$$V_{\text{shell}} = \pi r^2 dx$$

$$A_{\text{lateral}} = 2\pi r dx$$

Balance eqn:

$$in - out - gen - consumption = 0$$

Steady state

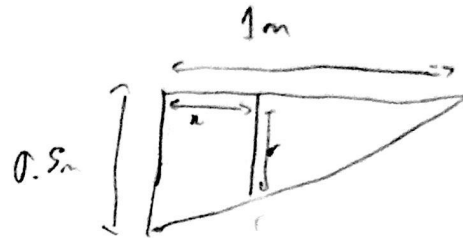
$$in = q|_x$$

$$r = r(x)$$

$$out = q|_{x+dx}$$

$$gen = \dot{q} V = \dot{q} \pi r^2 dx$$

$$cons = q'' A_c = q'' 2\pi r dx$$



Similar triangle identity: $\frac{r}{1-x} = \frac{0.5}{1} \Rightarrow r = \frac{1}{2}(1-x)$

Balance eqn:

B.C: $q(x=1)=0$
 the tip, $r=0$
 $\downarrow$
 $A_{lateral}=0$

$$q|_x - q|_{x+dx} + \dot{q} \pi r^2 dx - q'' 2\pi r dx = 0 \quad x \in [0, 1]$$

$$\frac{q|_x - q|_{x+dx}}{dx} + \dot{q} \pi r^2 - q'' 2\pi r$$

$$\Rightarrow \frac{-dq}{dx} + 100\pi \frac{1}{4}(1-x)^2 - 0.1 \times 2\pi \times \frac{1}{2}(1-x) = 0$$

$$-\frac{dq}{dx} + 25\pi(1-x^2) - 0.1\pi(1-x) = 0$$

$$\frac{dq}{dx} = \frac{\pi}{10}(1-x) - 25\pi(x^2 - 2x + 1)$$

$$\frac{dq}{dx} = \frac{\pi}{10} - \frac{\pi}{10}x - 25\pi x^2 + 50\pi x - 25\pi$$

$$\frac{dq}{dx} = -25\pi x^2 + \frac{499\pi}{10}x - \frac{249\pi}{10}$$

$$\frac{dq}{dx} = 25\pi x^2 - \frac{499\pi}{10}x + \frac{249\pi}{10}$$

$$q = \frac{25\pi}{3}x^3 - \frac{499\pi}{20}x^2 + \frac{249\pi}{10}x + A$$

$$B.C.: q(2,1) = 0 \Rightarrow 0 = \frac{25\pi}{3} - \frac{499\pi}{20} + \frac{249\pi}{10} + A$$

$$\Rightarrow A = -\frac{25\pi}{3} + \frac{499\pi}{20} - \frac{249\pi}{10} = -\frac{497\pi}{60}$$

$$\Rightarrow q = \frac{25\pi}{3} x^3 - \frac{499\pi}{20} x^2 + \frac{249\pi}{10} x - \frac{497\pi}{60}$$

$$q = -k \frac{dT}{dx} \pi r^2$$

$$\Rightarrow \frac{dT}{dx} = \frac{-1}{\pi k r^2} q$$

$$B.C.: T(x=0) = 30$$

$$x \in [0, 1]$$

$$\Rightarrow \frac{dT}{dx} = \frac{-4}{20\pi(1-x)^2} \left(\frac{25\pi}{3} x^3 - \frac{499\pi}{20} x^2 + \frac{249\pi}{10} x - \frac{497\pi}{60} \right)$$

$$\frac{dT}{dx} = \frac{-1}{5(1-x)^2} \left(\frac{25}{3} x^3 - \frac{499}{20} x^2 + \frac{249}{10} x - \frac{497}{60} \right)$$

$$T(x) = - \left[\frac{5}{3} \left(3 \ln|x-1| + \frac{x^3 + 3x^2 - 6}{2(x-1)} \right) - \frac{499}{100} \left(2 \ln|x-1| + \frac{x^2 - 2}{x-1} \right) + \frac{249}{10} \left(\ln|x-1| - \frac{1}{x-1} \right) + \frac{497}{60} \frac{1}{1-x} \right] + B$$

$$T(x) = \frac{-x(250x - 497)}{300} + B$$

$$B.C.: T(x=0) = 30^\circ C = 303K$$

$$\Rightarrow 303 = 0 + B \Rightarrow B = 303$$

$$\Rightarrow T(x) = \frac{x(497 - 250x)}{300} + 303$$

$$\Rightarrow T(x=1) = \frac{497 - 250}{300} + 303 = \boxed{606.2K}$$